

NILAKSHI ROY

Data Scientist

E: roynilakshiwork@gmail.com | Ph: +1 (669) 294-9152 | San Jose, CA (95134) | [linkedin.com/in/nilakshiroy](https://www.linkedin.com/in/nilakshiroy)

Portfolio: roynilakshi91.github.io/Nilakshi_new_portfolio

SUMMARY

Data Science professional with 5+ years of experience in advanced analytics, machine learning, and business intelligence. Skilled at analyzing large-scale datasets, building pipelines, and creating interactive BI dashboards that turn insights into business decisions. Hands-on expertise in predictive modeling, experimentation, KPI development, and risk analytics, with experience implementing agentic AI and process automation to improve operational efficiency.

WORK EXPERIENCE

Data Science and Analytics | TikTok USDS, San Jose, CA

[Aug 2022 - Present]

- o Conducted exploratory data analysis across large-scale datasets to uncover patterns and anomalies; built end-to-end data pipelines and **20+ interactive dashboards** with real-time alerts for metrics monitoring, ensuring data quality.
- o Built behavioral **predictive models** using gradient boosting (**CatBoost**) on product usage, engagement, and account-level signals, integrating model outputs into Customer Success workflows; validated through **A/B testing** and champion-challenger frameworks, achieving **98% precision** with a low false-positive rate, **reducing user feedback volume from 1,400 to 690/day** and **improving user experience**, exceeding quarterly goals.
- o Built **agentic AI workflows** using **Claude and MCP** to automate data retrieval, weekly performance analysis, metric monitoring, trend identification, and report generation, **improving weekly time efficiency by 86%**.
- o Conducted customer segmentation and cohort analyses to identify user behavior patterns; applied **causal inference** and experimentation to evaluate feature impact, contributing to a **~20% increase in global creator participation** in U.S. campaigns and a **94% success rate**.
- o Developed a device trust layer that **increased revenue by \$11K/day**, applying **TF-IDF text vectorization with gradient boosting** to analyze text from user platform activity and generate confidence scores.
- o **Prevented \$1.7M in capital losses** by developing perception models leveraging **LLMs** to identify abnormal content and detect coordinated cross-regional activity.

Data Specialist | Google, Mountain View, CA

Contract [May 2021 - Aug 2022]

- o Improved reporting dashboards and planning tools, **reducing data processing time by 80%**, and automated Weekly Business Reports (WBRs) via Apps Script & Excel macros, **cutting weekly man-hours by 25%**.
- o Managed opportunity tracking, data integrity and account mapping in Salesforce; built **SQL** test scripts for automated **ETL pipelines**, analyzing KPIs across sales trackers (GCP credit, Hub briefs, Deal trackers).
- o Partnered with the GCP Sales operations team on KPI deep-dives, delivering actionable insights to support GTM and sales strategy.

Data Reporting and Testing Analyst | Apple Inc., Sunnyvale, CA

Contract [May 2020 - Mar 2021]

- o Supported the EFA team, analyzing failure analysis and pre-/post-release reports for iPhone, iPad and Apple Watch iOS systems; extracted server data via queries to build flowcharts of key dimensions and metrics.
- o Tracked bug reports via Radar and FileMaker with cross-functional triage teams, and built **interactive Tableau dashboards** summarizing reports for daily executive stand-ups.

Data Analyst | American International Group (AIG), Bengaluru, India

[Sep 2016 - Apr 2019]

- o Analyzed large datasets to identify patterns and calculate potential losses from **natural calamities**, using MS Access, advanced Excel and **SQL** (Linux).
- o Built **Tableau dashboards** and data visualizations to make risk patterns understandable and presentable to stakeholders.
- o Communicated with clients to understand requirements and deliver within timelines; mentored juniors and new hires.

SKILLS

- o **Programming:** Python (Pandas, NumPy, SciPy), SQL, Spark SQL
- o **Machine Learning & Statistics:** Gradient Boosting (XGBoost, CatBoost), TF-IDF Vectorization, A/B Testing, Guardrail & North Star Metrics
- o **AI & Automation:** Agentic AI, Claude, MCP
- o **Analytics & Visualization:** EDA, ETL, Hive, Interactive BI Dashboards, Tableau, Matplotlib, Seaborn, MS Excel

EDUCATION

- o Master of Science in Computer Science | California State University, East Bay, Hayward, CA [02/2020 - 12/2021]
- o Master of Technology in Computer Science | Christ University, Bengaluru, India [08/2014 - 01/2016]

ACCOMPLISHMENTS

- o **Employee Recognition Award:** Top-quality analytics deliverables at AIG Analytics and Services.
- o **Top Performer Award:** exceeded quarterly goals at TikTok, improving detection accuracy and user experience.